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The effect of non-invasive physical interventions for accelerating tooth movement on orthodontically induced inflammatory root resorption: a systematic review and network meta-analysis

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Abstract

Objective Root resorption is one of the most common complications following orthodontic treatment, and severe cases may compromise tooth stability and function. Over the past decades, pharmacological, surgical, and non-invasive physical interventions have been explored to mitigate orthodontically induced root resorption. Among these approaches, non-invasive physical interventions have attracted increasing attention because of their favorable safety profile and clinical convenience. Therefore, this study aimed to evaluate the effects of non-invasive physical interventions on orthodontically induced inflammatory root resorption (OIIRR) using a network meta-analysis, and to identify the most effective intervention in order to provide evidence-based recommendations for clinical practice.

Methods A systematic search was conducted across *seven electronic databases* from their inception to identify studies investigating the effects of representative physical interventions on OIIRR. Study selection was performed in accordance with the Cochrane System Evaluator's Manual for Systematic Reviews. Following the PRISMA guidelines, statistical heterogeneity among studies was evaluated using the I^2 statistic. A random-effects model was applied when $I^2 > 50\%$. Network meta-analysis was performed using Stata 16.0. Forest plots were generated using RevMan 5.4.

Results A total of 1,135 records were screened, focusing on interventions including low-level laser therapy (LLLT), light-emitting diode (LED) therapy, and vibration. Twelve studies reporting OIIRR as an outcome measure were ultimately included. The results demonstrated that LED therapy significantly reduced OIIRR following orthodontic treatment (MD = −0.14 mm^3, 95% CI [−0.18, −0.10], $p < 0.00001$). Although LLLT (SMD = −0.31, 95% CI [−0.77, 0.14], $p = 0.17$) and vibration (SMD = −1.67, 95% CI [−3.54, 0.20], $p = 0.08$) showed a tendency toward reduced root resorption, these effects were not statistically significant. According to the SUCRA ranking, LED achieved a SUCRA value of 75.9 and a mean rank of 1.7, indicating a relative advantage over the other interventions in reducing OIIRR.

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Conclusions LED therapy appears to be effective in alleviating orthodontically induced inflammatory root resorption. However, although the reduction in OIIRR was statistically significant, whether the magnitude of this effect translates into clinically meaningful benefits remains uncertain. Further well-designed studies are warranted to determine optimal device parameters and application protocols to maximize therapeutic efficacy.

Keywords LED, LLLT, vibration, OIIRR

Introduction

Orthodontic treatment typically lasts for approximately two years or even longer [1]. Prolonged treatment duration has been associated with an increased risk of several undesirable complications, including root resorption, alveolar bone loss, and white spot lesions [2]. In recent years, considerable efforts have been devoted to accelerating orthodontic tooth movement in order to improve treatment efficiency and potentially reduce the incidence of orthodontic complications. Currently, approaches to accelerate tooth movement can be broadly categorized into surgical interventions, non-invasive physical interventions, and pharmacological strategies [3]. However, the clinical application of surgical techniques is limited by their invasive nature [4], while pharmacological approaches raise concerns regarding systemic side effects [5]. In particular, surgical procedures such as corticotomy accelerate tooth movement through the regional acceleratory phenomenon (RAP) [6]; nevertheless, whether this accelerated remodeling process exacerbates the severity of root resorption remains controversial [7].

Orthodontically induced inflammatory root resorption (OIIRR) is a common complication characterized by a reduction in root length and/or volume during orthodontic treatment [8]. During orthodontic tooth movement, periodontal tissues on the pressure side are subjected to mechanical compression, resulting in local ischemia and hypoxia. This microenvironment promotes anaerobic metabolism in periodontal cells and leads to tissue acidification [9]. These events trigger a cascade of acute, sterile inflammatory responses, which constitute the physiological basis of orthodontic tooth movement [10]. Mechanical loading frequently induces hyalinization of the periodontal ligament on the pressure side [11], a form of tissue necrosis that is largely unavoidable. Previous studies have demonstrated that OIIRR is closely associated with the removal of necrotic tissue induced by orthodontic forces [12]. Alarmingly, more than 91% of orthodontic patients exhibit some degree of OIIRR after treatment [13], with approximately 4% of severe cases showing apical root resorption exceeding 4 mm [14, 15]. Consequently, OIIRR is increasingly regarded as an inevitable and unpredictable adverse effect of orthodontic therapy [16]. The reduction in the root-to-crown ratio compromises tooth stability, and severe root resorption may ultimately increase the risk of tooth mobility or even tooth loss [17].

Non-invasive physical interventions, including photobiomodulation (PBM) and vibration, have attracted growing attention due to their convenience and favorable safety profiles. Compared with surgical and pharmacological approaches, these modalities are associated with reduced tissue trauma and minimal systemic adverse effects. However, whether such interventions increase the risk of root resorption while accelerating tooth movement remains a subject of ongoing debate. Importantly, most existing systematic reviews in this field predominantly include animal studies [18], and direct extrapolation of these findings to clinical orthodontic practice in humans should be approached with caution. Moreover, the aggressive marketing of non-surgical physical intervention devices by commercial manufacturers has intensified in recent years [19]. Under these circumstances, clinicians require robust, evidence-based evaluations of the expected effects of these interventions on OIIRR to avoid uncritical adoption and to support informed clinical decision-making. In light of these considerations, the present systematic review was conducted to evaluate the clinical effects of intraoral non-surgical, non-pharmacological adjunctive therapies on OIIRR.

Prior reviews have predominantly treated PBM as an undifferentiated category, failing to distinguish between specific modalities like LLLT and LED therapy, despite their differences in key physical properties (e.g., coherence, wavelength distribution, and energy delivery). These variations have the potential to influence both biological responses and clinical outcomes [20]. In previous studies, low-level laser therapy (LLLT) was defined as the application of coherent and monochromatic light generated by low-power laser equipment, while light-emitting diode (LED) therapy was defined as the use of non coherent broadband light sources [21]. Moreover, even when some reviews included a broader range of physical interventions (e.g., vibrational or pulsed ultrasound therapies), they did not perform direct or indirect comparisons between them. Compounding this issue, conventional pairwise meta-analysis is inherently unable to simultaneously compare and rank multiple competing interventions [19]. Consequently, the present study aims to address these gaps by distinguishing between different types of non-invasive physical interventions and applying a network meta-analysis to compare and rank their relative effectiveness in reducing orthodontically induced inflammatory root resorption.

Method

Guidance and eligibility criteria

The reporting of this systematic review and network meta-analysis followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses extension for Network Meta-Analyses (PRISMA-NMA) statement. This statement provides standardized reporting guidelines to ensure the completeness and transparency of network meta-analyses (the completed PRISMA-NMA checklist is provided in Supplementary Material S1 Table). The study protocol was prospectively registered with the International Prospective Register of Systematic Reviews (PROSPERO; registration number CRD420251052397).

Information sources and search strategy

A comprehensive electronic literature search was performed across seven databases: PubMed, the Cochrane Library, Embase, Web of science, China National Knowledge Infrastructure (CNKI), the Wanfang Database, and the Chinese Biomedical Literature Service System (SinoMed). Searches were conducted from the inception of each database to March 1, 2026. Both English and Chinese language publications were eligible for inclusion. Medical Subject Headings (MeSH) terms and relevant keywords used in the search strategy included Orthodontically Induced Inflammatory Root Resorption (OIIRR), Low-Level Laser Therapy (LLLT), Light-Emitting Diode (LED), andvibration.The complete search strategies for all databases are provided in Supplementary Material S2, with the strategy for PubMed serving as an example in Table 1. Search terms were appropriately adapted for each database to ensure consistency and completeness of the search strategy.

Table 1 Strategies used for searching PubMed for eligible publications

Steps	Search
#1	"low level light therapy"[MeSH Terms] OR ("low level"[All Fields] AND "light"[All Fields] AND "therapy"[All Fields]) OR "low level light therapy"[All Fields] OR ("low"[All Fields] AND "level"[All Fields] AND "laser"[All Fields] AND "therapy"[All Fields]) OR "low level laser therapy"[All Fields] OR ("Light-Emitting"[All Fields] AND ("diode"[All Fields] OR "diodes"[All Fields])) OR ("vibrate"[All Fields] OR "vibrated"[All Fields] OR "vibrates"[All Fields] OR "vibrating"[All Fields] OR "vibration"[MeSH Terms] OR "vibration"[All Fields] OR "vibrations"[All Fields] OR "vibrational"[All Fields] OR "vibrator"[All Fields] OR "vibrators"[All Fields]))
#2	(("orthodontal"[All Fields] OR "orthodontic"[All Fields] OR "orthodontical"[All Fields] OR "orthodontically"[All Fields] OR "orthodontics"[MeSH Terms] OR "orthodontics"[All Fields]) AND ("induce"[All Fields] OR "induced"[All Fields] OR "inducer"[All Fields] OR "inducers"[All Fields] OR "induces"[All Fields] OR "inducibilities"[All Fields] OR "inducibility"[All Fields] OR "inducible"[All Fields] OR "inducing"[All Fields]) AND ("inflammatories"[All Fields] OR "inflammatory"[All Fields]) AND ("root resorption"[MeSH Terms] OR ("root"[All Fields] AND "resorption"[All Fields]) OR "root resorption"[All Fields]))
#3	#1 AND #2

Inclusion criteria

Randomized controlled trials (RCTs) were eligible for inclusion if they enrolled patients undergoing orthodontic treatment for the first time, irrespective of the type of orthodontic appliance, archwire model, magnitude of applied force, amount of tooth movement, or frequency of appliance activation. The experimental groups received non-invasive physical interventions, including low-level laser therapy (LLLT), light-emitting diode (LED) therapy, or vibration. Different laser types used within LLLT protocols were all considered eligible. Control groups received conventional orthodontic treatment without any additional adjunctive physical intervention. The primary outcome measure was the extent of OIIRR, assessed from the initiation of orthodontic treatment to the predefined measurement time point.

Exclusion criteria

Non-randomized controlled studies were excluded, as previous evidence has suggested that intervention effects observed in non-RCTs may be overestimated when compared with those derived from RCTs [22]. In addition, in vitro studies, animal studies, and studies involving patients who were not undergoing orthodontic treatment for the first time were excluded. Studies that did not involve non-invasive physical interventions, such as LLLT, LED therapy, or vibration, were also excluded. Publications that did not report OIIRR as an outcome measure or from which relevant data could not be extracted were excluded. Furthermore, studies published in languages other than English or Chinese were not considered eligible.

Study selection

Two trained reviewers (JX and YJ) independently screened the titles and abstracts of retrieved records according to the predefined inclusion criteria. Full-text articles of potentially eligible studies were subsequently assessed by the same reviewers. When full texts were unavailable or relevant data were incomplete, the corresponding authors were contacted via email for clarification or data retrieval. If no response was received after three consecutive contact attempts, or if the studies were confirmed to be inaccessible, the publications were excluded. EndNote X9 was used to remove duplicate records. Any disagreements between the two reviewers were resolved through consultation with a third independent reviewer (JT) to achieve consensus.

Quality assessment

Two reviewers (MZ and ZJ) independently assessed the risk of bias for the included studies using the Cochrane Risk of Bias tool for randomized trials (RoB 2) [23]. This tool evaluates five domains: the randomization process, deviations from the intended interventions, missing outcome data, measurement of the outcome, and selection of the reported result. An overall judgment of 'low risk,' 'some concerns,' or 'high risk' of bias was made for each study [24]. In addition, to evaluate the certainty of NMA, we used the CINeMA framework [25]. The assessment considered the following domains: within-study bias (risk of bias), reporting bias, indirectness, imprecision, heterogeneity, and incoherence. Any disagreements between the reviewers were resolved through discussion with a third reviewer (LY). We used the Review Manager software (RevMan, version 5.4) to generate the risk-of-bias summary and graph based on the RoB 2 assessments.

Data analysis

Statistical analyses were performed using Stata software (version 16.0; StataCorp LLC, College Station, TX, USA). The extent of OIIRR was treated as a continuous outcome variable. Effect sizes were expressed as mean differences (MDs) with corresponding standard deviations (SDs) and 95% confidence intervals (CIs).

Statistical heterogeneity among studies was assessed using the I^2 statistic and the chi-square test, with a significance level set at $p < 0.10$. An I^2 value greater than 50% was considered indicative of substantial heterogeneity. In the presence of significant heterogeneity, a random-effects model was applied. Sensitivity analyses were conducted using Stata v16.0 to evaluate the robustness of the pooled estimates. Subgroup analyses were performed using Review Manager software (RevMan version 5.4; The Cochrane Collaboration). Statistical significance was defined as $p < 0.05$.

Network meta-analysis was also conducted using Stata v16.0. The evidence network plots, contribution plots, and comparison-adjusted funnel plots were generated using the mvmeta command. Inconsistency within closed loops of the evidence network was assessed by calculating the inconsistency factor, defined as the difference between direct and indirect estimates. A Z-test was applied to evaluate inconsistency, with $p > 0.05$ indicating no significant inconsistency.

Publication bias and small-study effects for each intervention were assessed separately based on direct comparison data. Funnel plots were constructed using Egger's linear regression approach, which evaluates the relationship between effect estimates and their precision, expressed as the inverse of the standard error. A symmetrical funnel plot and a p value greater than 0.05 were considered indicative of the absence of publication bias or small-study effects.

The surface under the cumulative ranking curve (SUCRA) was calculated to rank the relative effectiveness of each intervention. SUCRA values range from 0 to 100, with higher values indicating a greater probability of an intervention being among the most effective. Additionally, the probability of being the best treatment (PrBest) was estimated, with values closer to 100 suggesting a higher likelihood of being the optimal intervention.

Results

Study selection

The initial search conducted by two independent researchers identified 1,135 potentially relevant studies across multiple databases: 193 from PubMed, 64 from the Cochrane Library, 5 from the Embase, 18 from Web of science, 757 from CNKI, 5 from CBM, 90 from Wan-Fang, and 3 additional records identified through manual searching. After removing 113 duplicates across the major databases, the titles and abstracts of the remaining studies were screened, resulting in the exclusion of 967 irrelevant articles. Full-text assessment of the remaining studies led to the exclusion of 43 articles, which were either in vitro experiments, non-RCTs, reviews, systematic reviews, meta-analyses, or lacked extractable data. Ultimately, 12 studies [26–37] met the inclusion criteria and were included in this review. The literature screening process is summarized in Fig. 1.

Study characteristics

All 12 included studies [26–37] assessed orthodontically induced OIIRR as an outcome measure following orthodontic treatment. Among these, six studies [26–32] evaluated the effect of low-level laser therapy (LLLT) on OIIRR, three studies [27, 33] investigated light-emitting diode (LED) therapy, and four studies [34–37] examined the effect of mechanical vibration. The detailed characteristics of the included studies are summarized in Table 2.

Quality assessment

The detailed characteristics of the included studies are presented in Table 3. Of the 12 included studies, two were assessed as having a low risk of bias [26, 31], one was judged to have an unclear risk of bias [34], and nine were classified as having a high risk of bias [27–30, 32, 33, 35–37]. The overall risk-of-bias assessment is illustrated in Fig. 2A and B. An evaluation of the quality of evidence using the grade framework and the cinema tool showed that the quality of evidence for all comparisons was low or very low. The direct comparison with the control group (con vs. led, con vs. LLLT, con vs. vibration) was rated as extremely low due to significant within study bias and inconsistency. Indirect comparisons between

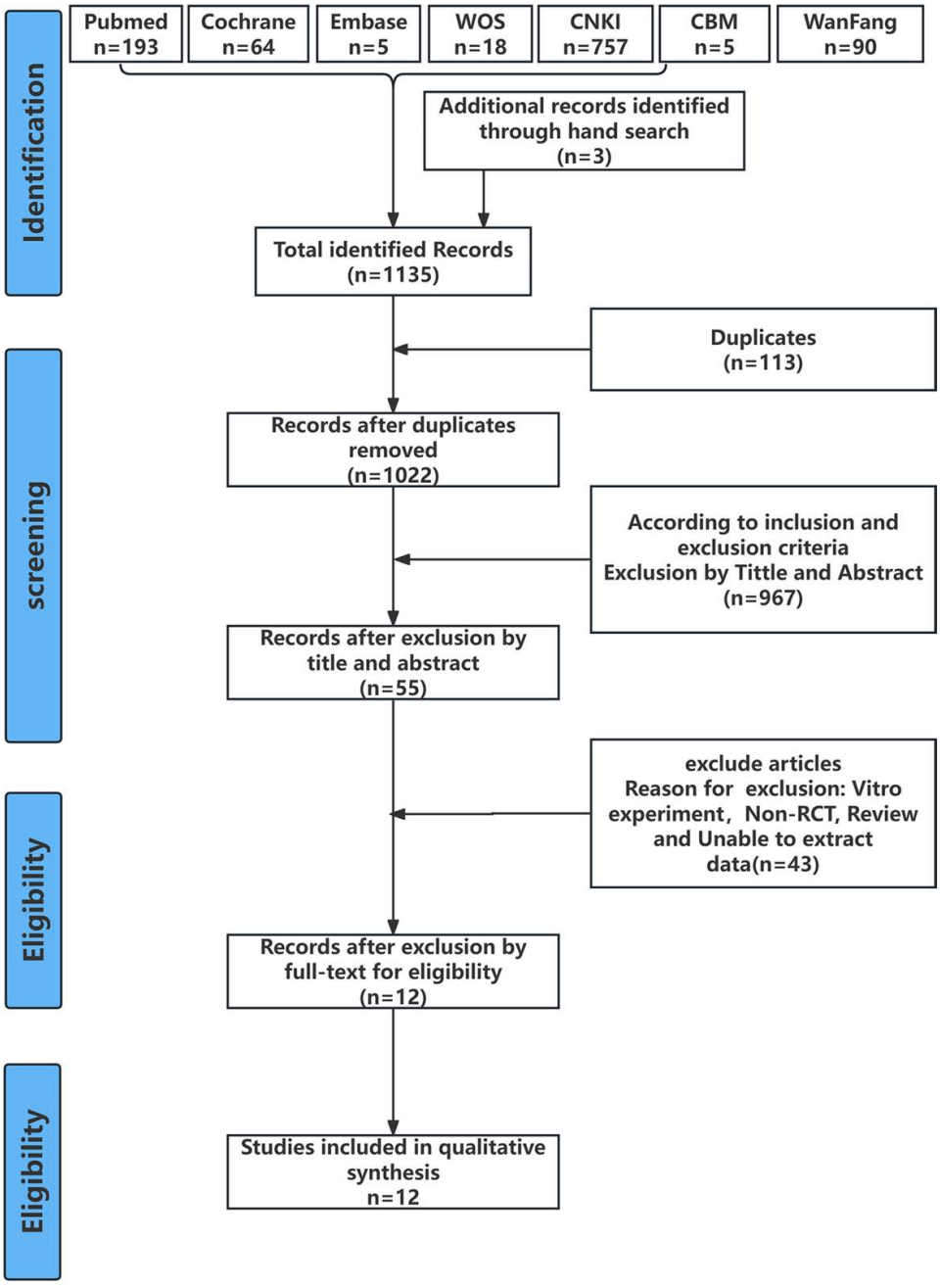


Fig. 1 Flow chart of study selection

interventions (LED vs. LLLT, led vs. vibration, LLLT vs. vibration) were rated as low in evidence quality due to inconsistencies in the network (Fig. 3).

Effect of interventions on OIIRR after orthodontic treatment

LLLT

With respect to the use of LLLT, seven studies [26–32] investigated the effect of LLLT on OIIRR following orthodontic treatment, with a total sample size of 240 participants. Due to substantial heterogeneity among the studies (I^2 = 66%), a random-effects model was applied. Since study [26, 27, 30, 31] used volume (mm^3) as the outcome measure, while studies [28, 29, 32] used root length (mm), the standardized mean difference (SMD) was employed for pooling the effect size. Consequently, the type of measurement unit (volume vs. length) served as the basis for subgroup analysis (Supplementary Material S3). The meta-analysis demonstrated that LLLT did not significantly reduce OIIRR after orthodontic treatment (SMD = – 0.31, 95% CI [– 0.77, 0.14], p = 0.17) (Table 4; Fig. 4). Therefore, from a statistical perspective,

Table 2 Characteristics of included studies

Study [27–38]	Country	Sample size	Intervention measures	Outcome measures (OIIRR)
Khaw et al. 2018 [27]	Australia	E = 20; C = 20	E: LLLT; C: Placebo	E: 0.746 ± 0.41/C: 0.779 ± 0.374
Goymen 2020 [28]	Turkey	E = 10; C = 10	E: LLLT/LED; C: Placebo	E (LLLT): 0.42 ± 0.07/C: 0.4 ± 0.06 E (LED): 0.25 ± 0.03/C: 0.4 ± 0.06
Hasan et al. 2020 [29]	Egypt	E = 15; C = 15	E: LLLT; C: Control	E: 0.69 ± 0.38/C: 0.74 ± 0.74
Joseph et al. 2022 [30]	India	E = 11; C = 11	E: LLLT; C: Control	E: 1.85 ± 1.433/C: 2.6 ± 1.619
Nayyer et al. 2021 [31]	India	E = 22; C = 22	E: LLLT; C: Control	E: 7.36 ± 2.15/C: 10.95 ± 3.95
Ng et al. 2018 [32]	Australia	E = 20; C = 20	E: LLLT; C: Placebo	E: 0.38 ± 0.16/C: 0.5 ± 0.16
Okla et al. 2018 [33]	UAE	E = 21; C = 23	E: LED; C: Control	E: 0.19 ± 0.63/C: 0.45 ± 1.07
Sambevski et al. 2022 [34]	Australia	E = 20; C = 20	E: LED; C: Control	E: 0.216 ± 0.229/C: 0.284 ± 0.259
DiBiase et al. 2016 [35]	UK	E = 27; C = 22	E: Vibration; C: Sham	E: 1.09 ± 0.64/C: 1.16 ± 0.94
El-Bialy et al. 2020 [36]	Canada	E = 13; C = 13	E: Vibration; C: Control	E: 0.0092 ± 0.0226/C: 0.0241 ± 0.0223
Raza et al. 2016 [37]	Canada	E = 20; C = 20	E: Vibration; C: Control	E: 0.48 ± 0.059/C: 1.01 ± 0.092
Yilmaz et al. 2021 [38]	Turkey	E = 10; C = 10	E: Vibration; C: Control	E: 0.476 ± 0.236/C: 0.462 ± 0.337

E Experimental group, *C* Control group, *OIIRR* Orthodontically induced inflammatory root resorption, *LLLT* Low-level laser therapy, *LED* Light emitting diod

LLLT intervention was not associated with a significant reduction in OIIRR following orthodontic treatment.

LED

Three studies [27, 33] evaluating the effect of LED therapy on OIIRR were included, with a total sample size of 60 participants. Given the low heterogeneity among the studies (I^2 = 4%), a fixed-effects model was applied. All included studies used resorption volume (mm^3) as the outcome measure; therefore, the mean difference (MD) was applied for the meta-analysis. Because statistical heterogeneity was low, subgroup analyses were not performed. The meta-analysis demonstrated a statistically significant reduction in OIIRR following orthodontic treatment in the LED group compared with the control group (MD = − 0.14, 95% CI [− 0.18, − 0.10], $p < 0.00001$) (Table 4; Fig. 5). These findings indicate that LED therapy may be effective in reducing the risk of OIIRR after orthodontic treatment.

Vibration

Four studies [34–37] investigated the effect of vibration on OIIRR following orthodontic treatment, with a total sample size of 135 participants. Owing to substantial heterogeneity among the studies (I^2 = 95%), a random-effects model was applied. Since study [36, 37] used volume (mm^3) as the outcome measure, while studies [34, 35] used root length (mm), the standardized mean difference (SMD) was employed for pooling the effect size. Consequently, the type of measurement unit (volume vs. length) served as the basis for subgroup analysis (Supplementary Material S4). The meta-analysis showed no statistically significant difference between the vibration and control groups (MD = − 1.67, 95% CI [− 3.54, 0.20], p = 0.08) (Table 4; Fig. 6). Therefore, based on the current statistical evidence, vibration intervention cannot be considered to significantly reduce OIIRR after orthodontic treatment.

Network meta-analysis

Figure 7 illustrates the structure of the evidence network. Four interventions were included in the network meta-analysis. Among them, four comparisons were based on direct evidence, while two comparisons were generated through indirect evidence derived from the network meta-analysis, namely vibration versus LLLT and vibration versus LED. The control group had the largest sample size, followed by the LLLT group. Given the limited number of nodes included in the network and the fact that direct comparisons between interventions were only available for LLLT and LED, the findings of this study should be interpreted with appropriate caution.

Following the application of physical interventions, the ranking plot (Fig. 8) and ranking results (Table 5) indicated that the LED group had the highest SUCRA value (75.9), followed by the vibration group (63.7), while the control group had the lowest SUCRA value (11.7). Similarly, the PrBest value was highest in the LED group (53.6), followed by the vibration group (34.3), and was lowest in the control group (0.2). The mean ranking values showed that the LED therapy ranked highest, with an ranking of 1.7.

However, although the SUCRA values of vibration and LED interventions were comparable, vibration did not significantly reduce OIIRR in the pairwise meta-analysis. These findings suggest that LED therapy may be more likely to represent the most effective intervention for alleviating OIIRR during orthodontic treatment.

Sensitivity analysis

A leave-one-out sensitivity analysis was performed to assess the robustness of the pooled estimates for OIIRR following treatment. Each study was sequentially

Table 3 Implementation details of the studies

Study [27–38]	Treatment	Age (years)	Intervention details
Khaw et al. 2018 [27]	Place a 0.017 × 0.025-inch beta-titanium cantilever on the first premolar and first molar, which have been bonded with SPEED self-ligating brackets, to apply a 150 g buccal tipping force for 4 weeks	13.8–18.7	aluminum-gallium-indium-phosphorus laser 75mW, 0.245 W·cm^{-2}, 3.6 J/cm^2, 15 s × 8 = 120 s
Goymen 2020 [28]	Place a 0.017 × 0.025-inch β-titanium cantilever on the first premolar and first molar, which have been bonded with MBT brackets (Mini Master Series metallic brackets), to apply a 150 g buccal tipping force	15-18.3	LLLT: GaAlAs laser 810 nm, 8 J·cm^{-2} (0, 3, 7, 14, 21, 28 days); LED: 850 nm, 850 nm^2, 10 min per day
Hasan et al. 2020 [29]	After placement of fixed orthodontic appliance, leveling and alignment till 16 × 22 mil stainless steel arch wire	18–25	Semiconductor diode laser 2.5 W for 120s applied only buccally at the middle third of the canine root in 1, 3, 4 days of canine retraction and then on every 2 weeks until complete canine retraction on one side was achieved
Joseph et al. 2022 [30]	Preadjusted edgewise appliance with McLaughlin, Bennet and Trevisi (MBT) prescription with 0.022-inch slot was bonded. Enmasse retraction was initiated after 21 days of engaging working 0.019 × 0.025-inch stainless steel archwires	18–30	Diode laser 975 nm, 4 J/cm^2, 0, 7 and 14 in the first month. Every 15 days for next three months on the experimental side
Nayyer et al. 2021 [31]	The first premolars were subjected to 150 g buccal force using a 0.019 × 0.025 inch beta-titanium alloy cantilever spring assembly extending from first molar to first premolar	13–30	InGaAs diode laser 980 nm, 100mW, 0, 3, 7, 11, 15 and 28 days after force application
Ng et al. 2018 [32]	150 g buccal-tipping forces were applied to maxillary first premolars over a 4-week period	Males:16.4 ± 1.3years females:16.7 ± 1.1years	AlGaAs diode laser 0.18 W, 808 nm, 1.6 J per point, 9s for continuous mode and 4.5 s for pulsed mode
Okla et al. 2018 [33]	0.022″ slot Master Series MBT prescription brackets or 0.022 slot Clarity Advanced Ceramic MBT prescription brackets. Wire sequence: 0.016 inch HANT with steel ties, 0.019 × 0.025 inch HANT and 0.019 × 0.025 inch stainless steel	12–40	OrthoPulse® device 850 mW, 0.065 J·cm^{-2}, 5 min per-arch-per-day
Sambevski et al. 2022 [34]	Buccal tipping force of 150 g was applied to each maxillary first premolar using a 0.0175 × 0.025 Beta III titanium	11.5–21.5	OrthoPulse® device 850 mW, 60 mW/cm^2, 5 min/day
DiBiase et al. 2016 [35]	MBT prescription 0.022-in precoated Victory series. Wire sequence: 0.014 inch, 0.018 inch, 0.018 × 0.025 inch nickel-titanium, and 0.019 × 0.025-in stainless steel archwires	12–19	AcceleDent 30 Hz, 0.2 N, 20 min/day
El-Bialy et al. 2020 [36]	Canine retraction using a 0.017 inch × 0.025 inch Titanium-Molybdenum alloy (TMA) canine retraction T-loop that was constructed for each extraction site	Average 19.7 ± 6.63	LIPUS frequency of 1.5 MHz, a pulse repetition rate of 1 kHz, average output intensity of 30 mW/cm2, 20 min/day
Raza et al. 2016 [37]	A 15u twist was applied in the arch wire using 0.019 × 0.025 inch TMA in a 0.022 inch bracket system (Synergy R) that produced a buccal root torque of approximately 5 N/mm at the bracket level	12–35	LIPUS, 30 mW/cm^2, 20 min for 4 weeks
Yilmaz et al. 2021 [38]	A buccally directed force of 150 g produced by a 0.017 × 0.025 TMA cantilever spring was applied to the premolar teeth on both sides	15.08–18.58	Oral-B HummingBird device 50 Hzwas applied to the maxillary first premolar for 10 min/day for 12 weeks

LLLT Low-level laser therapy, *LED* Light emitting diod

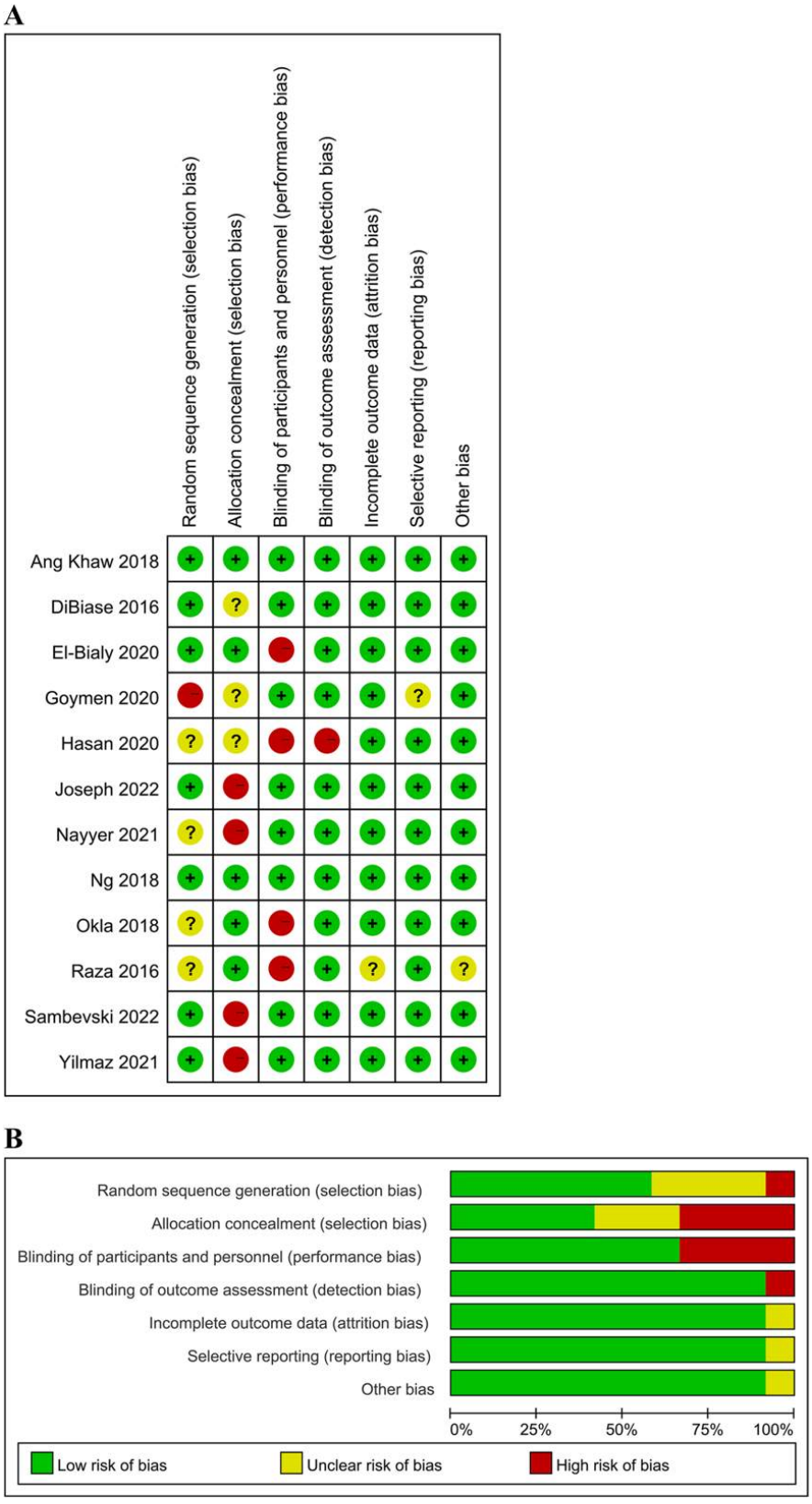


Fig. 2 **A** Risk of bias summary. **A** Risk of bias graph. **B** Risk of bias graph

Comparison	Number of Studies	Within-study bias	Reporting bias	Indirectness	Imprecision	Heterogeneity	Incoherence	Confidence rating
				Mixed evidence				
CON vs LED	3	Major concerns	Low risk	No concerns	No concerns	No concerns	Major concerns	Very low
CON vs LLLT	6	Some concerns	Low risk	No concerns	No concerns	Some concerns	Major concerns	Very low
CON vs Vibration	4	Major concerns	Low risk	No concerns	No concerns	No concerns	Major concerns	Very low
				Indirect evidence				
LED vs LLLT	--	Major concerns	Low risk	No concerns	No concerns	No concerns	Major concerns	Low
LED vs Vibration	--	Major concerns	Low risk	No concerns	No concerns	No concerns	Major concerns	Low
LLLT vs Vibration	--	Some concerns	Low risk	No concerns	No concerns	No concerns	Major concerns	Low

Fig. 3 Certainty of Evidence (GRADE) Assessment

Table 4 Meta-analysis results of different interventions

Interventions	Reference no	Heterogeneity		Effect model	Meta analysis results		
		I^2	*p*		Z	95% CI (%)	*p*
LLLT	[27–33]	66%	$p=0.007$	Random-effects model	-0.31	(-0.77, 0.14)	$P=0.17$
LED	[28, 34]	4%	$p=0.31$	Fixed-effects model	-0.14	(-0.18, -0.10)	$p<0.00001$
Vibration	[35–38]	95%	$p<0.00001$	Random-effects model	-1.67	(-3.54, 0.20)	$p=0.08$

LLLT Low-level laser therapy, *LED* Light emitting diode, *CI* Confidence interval

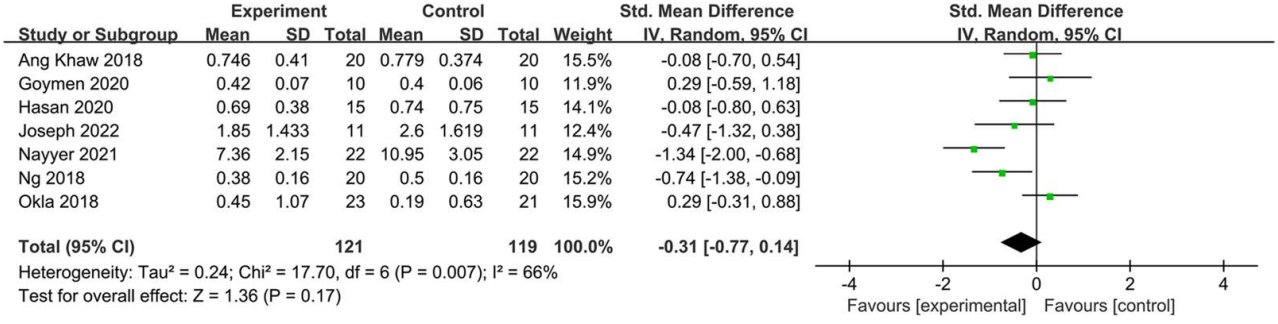


Fig. 4 Effect of LLLT on OIIRR control after orthodontic treatment

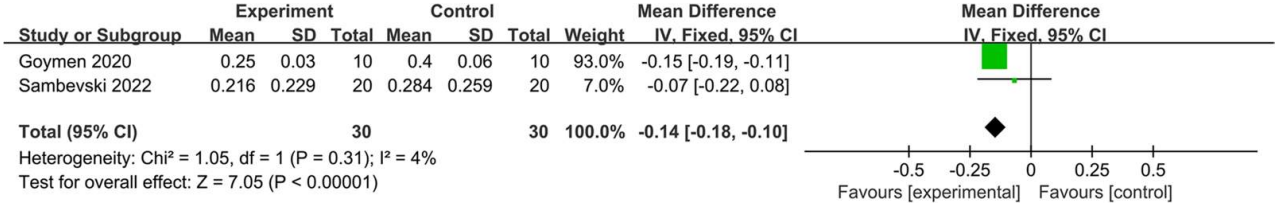


Fig. 5 Effect of LED on OIIRR control after orthodontic treatment

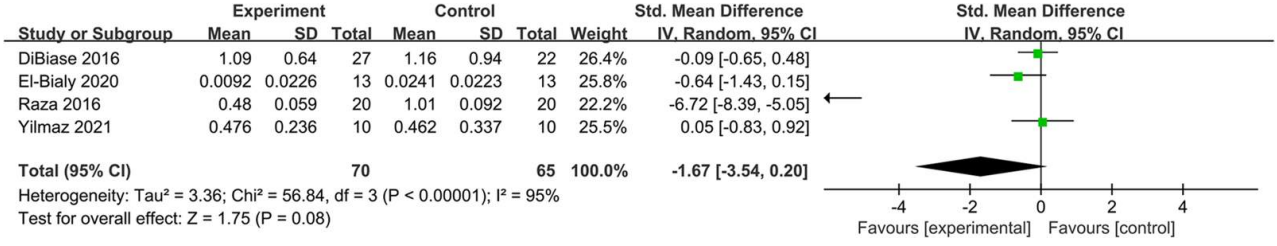


Fig. 6 Effect of vibration on OIIRR control after orthodontic treatment

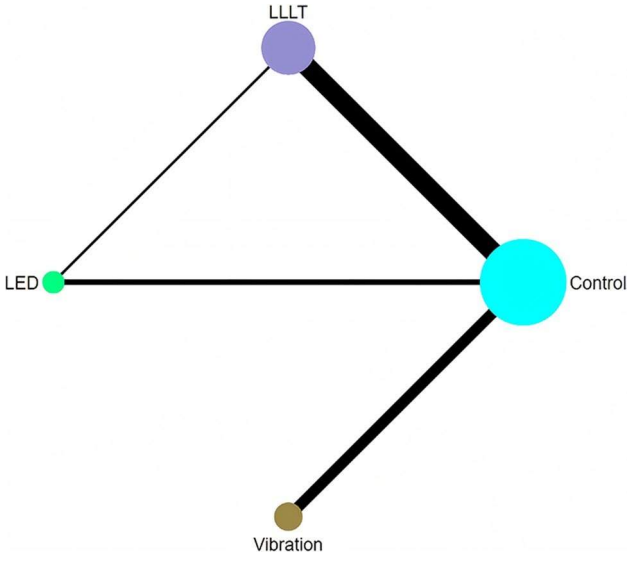


Fig. 7 Network graphs for various outcomes

excluded, and the pooled effect was recalculated. The sensitivity analysis showed that the overall results were not materially altered by the exclusion of any single study, indicating that the pooled estimates in this meta-analysis were stable and reliable. The results of the sensitivity analysis are presented in Fig. 9.

Table 5 Sorting results

Treatment	SUCRA	PrBest	MeanRank
Control	11.7	0.2	3.6
LLLT	48.7	11.8	2.5
LED	75.9	53.6	1.7
Vibration	63.7	34.3	2.1

Publication bias

Publication bias was assessed using Egger's linear regression test for the OIIRR outcomes. The analysis yielded a *p* value of 0.5494, which exceeded the predefined significance threshold of 0.05, indicating no evidence of significant publication bias in the included studies. The results are illustrated in Fig. 10.

Discussion

Accelerating orthodontic tooth movement has long been a central objective for orthodontists. Achieving faster tooth movement within a safe and controllable range may not only reduce the risk and severity of orthodontic complications but also improve patients' treatment experience [38]. Although the effectiveness of non-surgical and non-pharmacological physical interventions in accelerating tooth movement remains controversial, recent systematic reviews and meta-analyses have suggested that LED therapy [39], LLLT [40], and vibration [41] may effectively accelerate orthodontic tooth movement and shorten overall treatment duration. Current evidence indicates that osteoclasts involved in bone

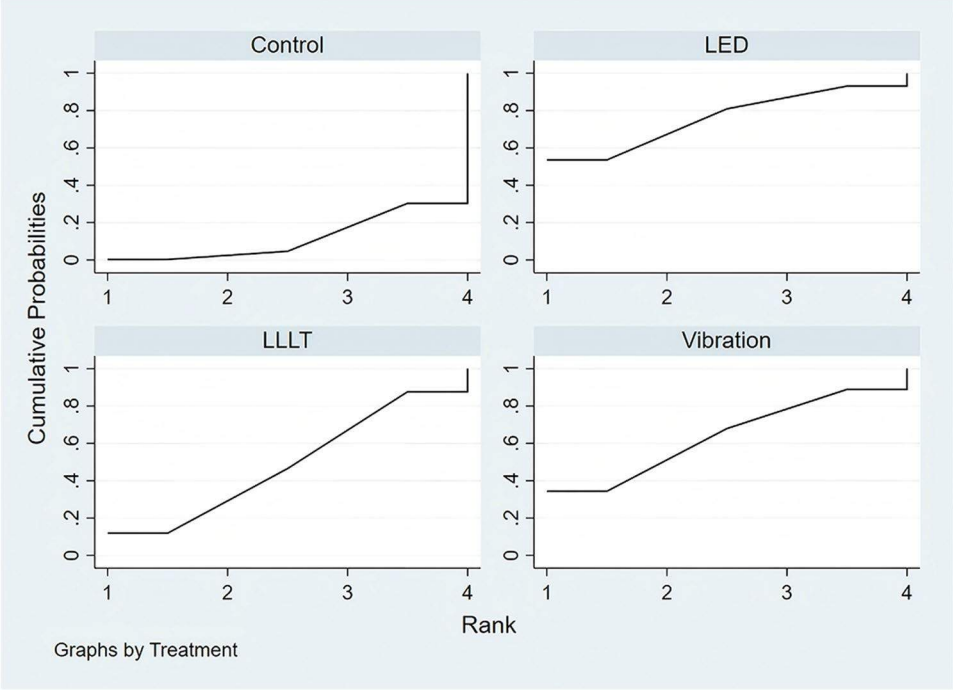


Fig. 8 SUCRA probability ranking. SUCRA, surface under the cumulative ranking curve

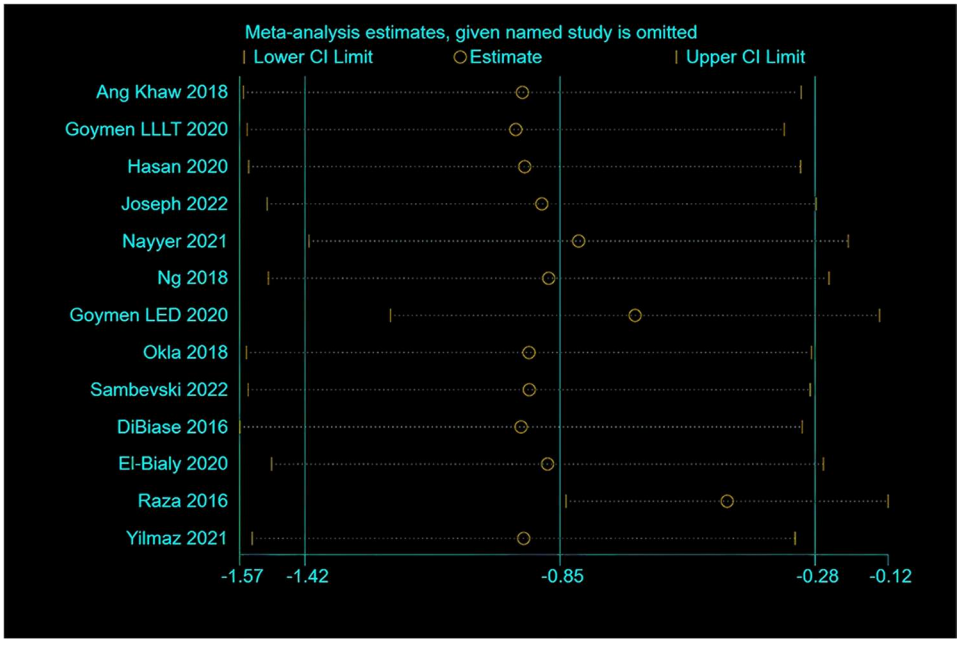


Fig. 9 Sensitivity analysis

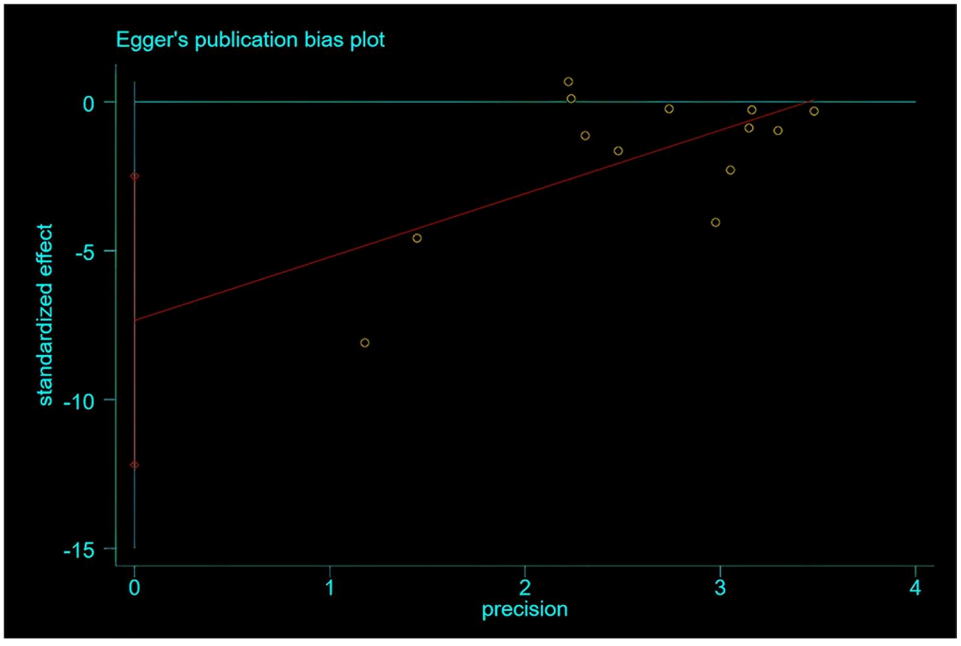


Fig. 10 Funnel plots for publication bias

remodeling represent the primary biological targets of these non-surgical interventions, largely through the upregulation of inflammatory processes [42]. However, osteoclast recruitment and activity at sites of root surface damage are also considered the principal contributors to orthodontically induced inflammatory root resorption (OIIRR) [16]. Interestingly, LED, LLLT, and vibration—representative physical interventions aimed at accelerating tooth movement—appear to simultaneously promote tooth movement while alleviating OIIRR, a phenomenon that seems paradoxical.

To elucidate this apparent clinical paradox, it is essential to consider the underlying physiological mechanisms of orthodontic tooth movement and OIIRR. The rate-limiting step of orthodontic tooth movement occurs on the pressure side [43], where compressive stress induces varying degrees of vascular occlusion and subsequent hyalinization. The recruitment of inflammatory cells to

these hyalinized areas is a prerequisite for effective tooth movement [44], and OIIRR typically occurs concurrently during the removal of hyalinized tissue by phagocytic cells [45]. When hyalinization persists, osteoclasts remain active around the tooth root. Once the integrity of the cementum is compromised and dentin becomes exposed, root resorption may progress rapidly and extensively [46]. Thus, the development of OIIRR is not solely determined by the number of osteoclasts but is more closely related to alterations in the local microenvironment induced by orthodontic forces. OIIRR is strongly associated with prolonged hyalinization, sustained ischemia, and persistent inflammation on the pressure side of the periodontal tissues. Delayed clearance of necrotic tissue may allow osteoclast-like cells to extend their activity onto the root surface, thereby initiating root resorption. This mechanism underlies the long-standing clinical recommendation to apply intermittent orthodontic forces rather than continuous forces [47]. The auxiliary interventions included in the present study appear to facilitate cellular activation, shorten cell cycle duration, and accelerate bone remodeling, thereby enabling accelerated tooth movement while potentially mitigating the severity of root resorption.

LED and LLLT both belong to the category of photobiomodulation (PBM) [48], which can be classified into class III (output power ≤ 500 mW) and class IV (output power > 500 mW) based on power output in periodontal tissues [49]. Accumulating evidence suggests that noncoherent LED light with parameters comparable to those of low-power coherent monochromatic lasers can elicit similar beneficial biological effects [50], while offering advantages in terms of cost-effectiveness, convenience, and safety [51]. Previous studies have shown that PBM does not significantly alter the OPG/RANKL ratio during orthodontic tooth movement [52], nor does it reduce the rate of tooth movement, suggesting that PBM may alleviate OIIRR through alternative mechanisms. The primary cellular target of PBM is cytochrome c oxidase (CCO) within the mitochondrial respiratory chain. Upon absorption of red or near-infrared light, mitochondrial function is enhanced, membrane potential increases, and electron transport is accelerated, leading to increased intracellular adenosine triphosphate (ATP) production [53]. PBM has also been shown to promote the generation of reactive oxygen species (ROS) in periodontal ligament stem cells, thereby stimulating cellular proliferation [54]. Following PBM irradiation, mitochondrial redox potential shifts toward an oxidative state, increasing ROS production and activating transcription factors such as nuclear factor-κB (NF-κB) and hypoxia-inducible factor-1 (HIF-1) [55], which in turn promote gene transcription and protein synthesis. Moreover, PBM can activate photosensitive ion channels, allowing calcium ions (Ca^{2+}) to enter cells and modulate multiple signaling pathways via ROS, cyclic adenosine monophosphate (cAMP), nitric oxide (NO), and Ca^{2+} release [56]. As a key second messenger, Ca^{2+} plays a critical role in periodontal ligament cell signal transduction, including responses to mechanical stimuli [57], thereby influencing cellular secretion, proliferation, growth, and differentiation [58].

Mechanical vibration in orthodontics refers to the application of periodic mechanical loads to teeth and surrounding supporting tissues, including the periodontal ligament and alveolar bone, to stimulate bone remodeling [59]. Vibration exerts biological stimulatory effects on both osteoblasts and osteoclasts and increases cell number and activity within the periodontal ligament, thereby contributing to alveolar bone remodeling [60]. These mechanical stimuli are perceived by cell membrane receptors such as integrins [61] and G protein–coupled receptors (GPCRs) [62], which activate multiple mechanotransduction pathways in bone cells. In osteoclasts, vibration has been shown to enhance bone resorption by increasing RANKL expression [63], whereas in osteoblasts, vibration promotes the expression of osteogenic markers including runt-related transcription factor 2 (RUNX2), osteoprotegerin (OPG) [64], and alkaline phosphatase (ALP). Although fluctuations in OPG and RANKL expression have been observed, the OPG/RANKL ratio does not appear to change significantly [65].

In the present study, a systematic review combined with network meta-analysis was conducted to evaluate the effects of multiple non-invasive physical interventions on OIIRR. From an initial pool of 1,135 studies, 12 studies reporting OIIRR outcomes were ultimately included. Pairwise meta-analysis, frequentist-based network meta-analysis, and SUCRA analyses were performed. The pairwise meta-analysis demonstrated that adjunctive LED therapy during orthodontic tooth movement significantly reduced the severity of OIIRR, whereas LLLT and vibration did not exhibit statistically significant protective effects. According to the SUCRA ranking, the likelihood of vibration benefits is slightly lower compared to LED; The average ranking values of LED and vibration are similar and significantly better than LLLT. Importantly, vibration did not show statistically significant efficacy in paired meta-analyses and was associated with significant heterogeneity. Therefore, when both effect estimates and statistical robustness are considered, LED therapy may represent the most reliable physical intervention for reducing OIIRR following orthodontic treatment.

In the present study, LED therapy was associated with a mean reduction of 0.14 mm in root resorption, a difference that was detectable radiographically and statistically significant. However, the clinical relevance of this magnitude of difference should be interpreted with caution. Previous studies have reported that the incidence of severe orthodontically induced inflammatory root

resorption (OIIRR), defined as root shortening greater than 4 mm, is relatively low, occurring in approximately 4% of patients [15]. Moreover, the clinical consequences of root resorption are influenced by cumulative effects and individual susceptibility. For many patients, a difference of 0.14 mm may have limited impact on clinical decision-making. Nevertheless, for individuals at higher risk of OIIRR or those presenting with relatively short roots prior to treatment, even a small reduction in root resorption may have a potentially protective effect [44]. It should therefore be emphasized that the findings of the present study primarily provide evidence at the population level, suggesting a potentially protective tendency associated with LED therapy. The clinical significance of this effect at the individual level remains uncertain and requires further confirmation through well-designed prospective studies with longer follow-up periods and clinically relevant outcomes, such as tooth survival and mobility.

The heterogeneity between LLLT and vibration in this study is large, and its source may be the significant differences in intervention parameters (such as wavelength, power, energy density of laser; equipment, frequency, and application duration of vibration). Furthermore, it should be acknowledged that the present study included only a limited number of intervention nodes, resulting in a relatively sparse network structure. This may reduce the robustness of treatment rankings and limit the interpretability of the SUCRA results; therefore, the ranking probabilities should be interpreted with caution. The certainty of evidence may be limited due to the high risk of bias observed in many of the included studies. According to the GRADE framework, methodological limitations such as lack of blinding and unclear allocation procedures may lead to downgrading of the certainty of evidence. Future large-scale, well-designed randomized controlled trials are required to provide more robust evidence and further confirm these findings.

Conclusion

This network meta-analysis suggests that LED therapy may reduce orthodontically induced inflammatory root resorption to some extent. However, the clinical significance of this finding remains uncertain and should be interpreted with caution. Further high-quality randomized controlled trials with standardized intervention parameters are warranted to clarify the role of non-invasive physical interventions in preventing OIIRR.

Abbreviations

OIIRR	Orthodontically induced inflammatory root resorption
LLLT	Low-level laser therapy
LED	Light-emitting diode
PBM	Photobiomodulation
PBM	Photobiomodulation
RAP	Regional acceleratory phenomenon
RCT	Randomized controlled trial
PRISMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses
PRISMA-NMA	Preferred Reporting Items for Systematic Reviews and Meta-Analyses for Network Meta-Analyses
CNKI	China National Knowledge Infrastructure
MeSH	Medical Subject Headings
MD	Mean difference
SD	Standard deviation
CI	Confidence interval
SUCRA	Surface under the cumulative ranking curve
PrBest	Probability of being the best treatment
RoB	Risk of bias
ATP	Adenosine triphosphate
CCO	Cytochrome c Oxidase
ROS	Reactive oxygen species
NF-κB	Nuclear factor-kappa B
HIF-1	Hypoxia-inducible factor-1
cAMP	Cyclic adenosine monophosphate
NO	Nitric oxide
Ca^{2+}	Calcium ion
GPCRs	G protein–coupled receptors
RUNX2	Runt-related transcription factor 2
OPG	Osteoprotegerin
RANKL	Receptor activator of nuclear factor-κB ligand
ALP	Alkaline phosphatase
LIPUS	Low-intensity pulsed ultrasound
TMA	Titanium-molybdenum alloy

Supplementary Information

The online version contains supplementary material available at https://doi.org/10.1186/s12903-026-08301-x.

Supplementary Material 1: S1 Table. The PRISMA network meta-analysis checklist.

Supplementary Material 2: S2 Table. Search strategy of each database.

Supplementary Material 3: S3 Fig. Subgroup analysis of LLLT.

Supplementary Material 4: S4 Fig. Subgroup analysis of vibration.

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Authors' contributions

Conceptualization and idea: JXL and JTH. Screening and extraction: JXL, YJL, XXX, MZC and ZJY. Original draft preparation: JXL and LYM. Review and editing: JXL and JTH. Supervision: JTH. All authors read and approved the final manuscript.

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Data availability

All data and materials are available from the corresponding author.

Declarations

Ethics approval and consent to participate

Not applicable.

Consent for publication
Not applicable.

Competing interests
The authors declare no competing interests.

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References

1. Khalil R, Sukhia RH, Fida M. The influence of orthodontist change on treatment duration and outcomes in patients treated with Clark's twin block appliance followed by non-extraction fixed mechanotherapy – a retrospective cohort study. BMC Oral Health. 2024;24:1521. https://doi.org/10.1186/s12903-024-05351-x.
2. Alam MK, Kanwal B, Abutayyem H, Alswairki HJ, Alfawzan AA, Shqaidef A, et al. Complications Arising Due to Orthodontic Treatment—A Systematic Review and Meta-Analysis. Appl Sci. 2023;13:4035. https://doi.org/10.3390/app13064035.
3. Kacprzak A, Strzecki A. Methods of accelerating orthodontic tooth movement:A review of contemporary literature. Dent Med Probl. 2018;55:197–206. https://doi.org/10.17219/dmp/90989.
4. Bisconti F, Eva M, Thevenet E, Zamora-Martinez N. Comparison of efficacy between micro-osteoperforations and alveolar corticotomies on the rate of orthodontic tooth movement: A systematic review and meta-analysis. J Cranio-Maxillofacial Surg. 2025;53:459–75. https://doi.org/10.1016/j.jcms.2025.01.013.
5. Eltimamy A, El-Sharaby FA, Eid FH, El-Dakrory AE. The Effect of Local Pharmacological Agents in Acceleration of Orthodontic Tooth Movement: A Systematic Review. Open Access Maced J Med Sci. 2019;7:882–6. https://doi.org/10.3889/oamjms.2019.203.
6. Keser E, Naini FB. Accelerated orthodontic tooth movement: surgical techniques and the regional acceleratory phenomenon. Maxillofac Plast Reconstr Surg. 2022;44:1. https://doi.org/10.1186/s40902-021-00331-5.
7. Sirri MR, Hajeer MY, Baba F, Aljabban O, Burhan AS, Namera MO. Root resorption associated with minimally invasive surgical acceleration of orthodontic tooth movement: a systematic review. Eur J Orthod. 2025;47:cjaf035. https://doi.org/10.1093/ejo/cjaf035.
8. Yu H, Wu Z, Bao X, Tang X, Zhang J, Zhang Y, et al. A sustained-release Trametinib bio-multifunction hydrogel inhibits orthodontically induced inflammatory root resorption. RSC Adv. 2022;12:16444–53. https://doi.org/10.1039/D2RA00763K.
9. Li B, Wang L, He H. Autophagy in orthodontic tooth movement: advances, challenges, and future perspectives. Mol Med. 2025;31:245. https://doi.org/10.1186/s10020-025-01299-y.
10. Osada A, Hitomi S, Nakajima A, Hayashi Y, Shibuta I, Tsuboi Y, et al. Periodontal acidification contributes to tooth pain hypersensitivity during orthodontic tooth movement. Neurosci Res. 2022;177:103–10. https://doi.org/10.1016/j.neures.2021.11.007.
11. Cuoghi OA, Topolski F, De Faria LP, Ervolino E, Micheletti KR, Miranda-Zamalloa YM, et al. Correlation between pain and hyalinization during tooth movement induced by different types of force. Angle Orthod. 2019;89:788–96. https://doi.org/10.2319/041118-273.1.
12. He W, Fu Y, Yao S, Huang L. Programmed cell death of periodontal ligament cells. J Cell Physiol. 2023;238:1768–87. https://doi.org/10.1002/jcp.31091.
13. Lund H, Gröndahl K, Hansen K, Gröndahl H-G. Apical root resorption during orthodontic treatment: A prospective study using cone beam CT. Angle Orthod. 2012;82:480–7. https://doi.org/10.2319/061311-390.1.
14. Weltman B, Vig KWL, Fields HW, Shanker S, Kaizar EE. Root resorption associated with orthodontic tooth movement: A systematic review. Am J Orthod Dentofac Orthop. 2010;137:462–76. https://doi.org/10.1016/j.ajodo.2009.06.021.
15. Tieu LD, Saltaji H, Normando D, Flores-Mir C. Radiologically determined orthodontically induced external apical root resorption in incisors after non-surgical orthodontic treatment of class II division 1 malocclusion: a systematic review. Prog Orthod. 2014;15:48. https://doi.org/10.1186/s40510-014-0048-7.
16. Yamaguchi M, Fukasawa S. Is Inflammation a Friend or Foe for Orthodontic Treatment? Inflammation in Orthodontically Induced Inflammatory Root Resorption and Accelerating Tooth Movement. IJMS. 2021;22:2388. https://doi.org/10.3390/ijms22052388.
17. Yang J, Zhang Q, Zheng D, Hou F. Photobiomodulation and orthodontic root resorption: A systematic review and meta-analysis. Photodiagn Photodyn Ther. 2025;54:104659. https://doi.org/10.1016/j.pdpdt.2025.104659.
18. Haugland L, Kristensen KD, Lie SA, Vandevska-Radunovic V. The effect of biologic factors and adjunctive therapies on orthodontically induced inflammatory root resorption: a systematic review and meta-analysis. Eur J Orthod. 2018;40:326–36. https://doi.org/10.1093/ejo/cjy003.
19. Al-Dboush R, Flores-Mir C, El-Bialy T. Impact of intraoral non-pharmacological non-surgical adjunctive interventions on orthodontically induced inflammatory root resorption in humans: A systematic review. Orthod Craniofac Res. 2021;24:459–79. https://doi.org/10.1111/ocr.12476.
20. Zhang G, Li Z, McGrath C, Yang Y, Shan Z. Effects of photobiomodulation therapy on orthodontically induced inflammatory root resorption: a systematic review and meta-analysis. BMC Oral Health. 2025;26:162. https://doi.org/10.1186/s12903-025-07512-y.
21. Heiskanen V, Hamblin MR. Photobiomodulation: lasers vs. light emitting diodes? Photochem Photobiol Sci. 2018;17:1003–17. https://doi.org/10.1039/c8pp00176f.
22. Papageorgiou SN, Xavier GM, Cobourne MT. Basic study design influences the results of orthodontic clinical investigations. J Clin Epidemiol. 2015;68:1512–22. https://doi.org/10.1016/j.jclinepi.2015.03.008.
23. Rong W, Geok SK, Samsudin S, Zhao Y, Ma H, Zhang X. Effects of strength training on neuromuscular adaptations in the development of maximal strength: a systematic review and meta-analysis. Sci Rep. 2025;15:19315. https://doi.org/10.1038/s41598-025-03070-z.
24. Luo J, Xu P, Shuai H, Cai T, Cui S, Zhou L, et al. Comparative analysis of the long-term efficacy and safety of minimally invasive simple prostatectomy and endoscopic enucleation of the prostate for large benign prostatic hyperplasia (> 80 mL). Minerva Urol Nephrol. 2025;76. https://doi.org/10.23736/S2724-6051.24.05940-8.
25. Birkinshaw H, Friedrich CM, Cole P, Eccleston C, Serfaty M, Stewart G et al. Antidepressants for pain management in adults with chronic pain: a network meta-analysis. Cochrane Database of Systematic Reviews. 2023;2023. https://doi.org/10.1002/14651858.CD014682.pub2
26. Ang Khaw CM, Dalci O, Foley M, Petocz P, Darendeliler MA, Papadopoulou AK. Physical properties of root cementum: Part 27. Effect of low-level laser therapy on the repair of orthodontically induced inflammatory root resorption: A double-blind, split-mouth, randomized controlled clinical trial. Am J Orthod Dentofac Orthop. 2018;154:326–36. https://doi.org/10.1016/j.ajodo.2018.04.022.
27. Goymen M, Gulec A. Effect of photobiomodulation therapies on the root resorption associated with orthodontic forces: a pilot study using micro computed tomography. Clin Oral Invest. 2020;24:1431–8. https://doi.org/10.1007/s00784-019-03155-w.
28. Mohamed Hasan N, Sabet N, El Ghoul D. The Effect Of Low-Level Laser Therapy On The Rate Of Tooth Movement During Maxillary Canine Retraction: A Randomized Clinical Trial. Ain Shams Dent J. 2021;21:105–11. https://doi.org/10.21608/asdj.2021.59455.1014.
29. Joseph A, Prashantha G, Sabrish S, Sagarkar R, Mathew S. Comparison of Rate of Tooth Movement, Root Resorption and Pulp Vitality during En masse Anterior Retraction with Micro-osteoperforation and Low Level Laser Therapy: A Randomised Clinical Trial. JCDR. 2022. https://doi.org/10.7860/JCDR/2022/55298.16802.
30. Nayyer N, Tripathi T, Rai P, Kanase A. Effect of photobiomodulation on external root resorption during orthodontic tooth movement – a randomized controlled trial. Int Orthod. 2021;19:197–206. https://doi.org/10.1016/j.ortho.2021.01.007.
31. Ng D, Chan AK, Papadopoulou AK, Dalci O, Petocz P, Darendeliler MA. The effect of low-level laser therapy on orthodontically induced root resorption: a

pilot double blind randomized controlled trial. Eur J Orthod. 2018;40:317–25. https://doi.org/10.1093/ejo/cjx065.
32. Okla NA, Bader DMA, Makki L. Effect of Photobiomodulation on Maxillary Decrowding and Root Resorption: A Randomized Clinical Trial. APOS. 2018;8:86–91. https://doi.org/10.4103/apos.apos_46_18.
33. Sambevski J, Papadopoulou AK, Foley M, Dalci K, Petocz P, Darendeliler MA, et al. Physical properties of root cementum: Part 29. The effects of LED-mediated photobiomodulation on orthodontically induced root resorption and pain: a pilot split-mouth randomized controlled trial. Eur J Orthod. 2022;44:650–8. https://doi.org/10.1093/ejo/cjac022.
34. DiBiase AT, Woodhouse NR, Papageorgiou SN, Johnson N, Slipper C, Grant J, et al. Effect of supplemental vibrational force on orthodontically induced inflammatory root resorption: A multicenter randomized clinical trial. Am J Orthod Dentofac Orthop. 2016;150:918–27. https://doi.org/10.1016/j.ajodo.2016.06.025.
35. El-Bialy T, Farouk K, Carlyle TD, Wiltshire W, Drummond R, Dumore T, et al. Effect of Low Intensity Pulsed Ultrasound (LIPUS) on Tooth Movement and Root Resorption: A Prospective Multi-Center Randomized Controlled Trial. JCM. 2020;9:804. https://doi.org/10.3390/jcm9030804.
36. Raza H, Major P, Dederich D, El-Bialy T. Effect of low-intensity pulsed ultrasound on orthodontically induced root resorption caused by torque: A prospective, double-blind, controlled clinical trial. Angle Orthod. 2016;86:550–7. https://doi.org/10.2319/081915-554.1.
37. Yilmaz H, Ozlu FC, Turk T, Darendeliler MA. The effect of 12 weeks of mechanical vibration on root resorption: a micro-CT study. Prog Orthod. 2021;22:28. https://doi.org/10.1186/s40510-021-00369-1.
38. Zheng Y, Lam XY, Wu M, Lin Y. Acceleration of Orthodontic Tooth Movement: A Bibliometric and Visual Analysis. Int Dent J. 2025;75:1223–33. https://doi.org/10.1016/j.identj.2024.08.012.
39. Huang T, Wang Z, Li J. Efficiency of photobiomodulation on accelerating the tooth movement in the alignment phase of orthodontic treatment—A systematic review and meta-analysis. Heliyon. 2023;9:e13220. https://doi.org/10.1016/j.heliyon.2023.e13220.
40. Bakdach WMM, Hadad R. Effectiveness of low-level laser therapy in accelerating the orthodontic tooth movement: A systematic review and meta-analysis. Dent Med Probl. 2020;57:73–94. https://doi.org/10.17219/dmp/112446.
41. Lalitwageewong T, Leethanakul C, Witeerungrot T. Effective mechanical vibration frequency in canine tooth movement acceleration: a systematic review and network meta-analysis of randomized controlled trial studies. Clin Oral Invest. 2025;29:292. https://doi.org/10.1007/s00784-025-06338-w.
42. El-Angbawi A, McIntyre G, Fleming P, Bearn D. Non-surgical adjunctive interventions for accelerating tooth movement in patients undergoing orthodontic treatment. Cochrane Database Syst Reviews. 2023;2023. https://doi.org/10.1002/14651858.CD010887.pub3.
43. Alansari S, Sangsuwon C, Vongthongleur T, Kwal R, Teo MC, Lee YB, et al. Biological principles behind accelerated tooth movement. Semin Orthod. 2015;21:151–61. https://doi.org/10.1053/j.sodo.2015.06.001.
44. Dawood HM, Kroeger A, Chavda V, Chapple ILC, Kebschull M. Under pressure—mechanisms and risk factors for orthodontically induced inflammatory root resorption: a systematic review. Eur J Orthod. 2023;45:612–26. https://doi.org/10.1093/ejo/cjad011.
45. Both A, Ibrahim G, Marciniak J, Rath-Deschner B, Küchler EC, Kirschneck C, et al. Bone, dentin and cementum differentially influence the differentiation of osteoclast-like cells. Sci Rep. 2025;15:19857. https://doi.org/10.1038/s41598-025-04874-9.
46. Patel S, Saberi N, Pimental T, Teng P. Present status and future directions: Root resorption. Int Endodontic J. 2022;55:892–921. https://doi.org/10.1111/iej.13715.
47. Ozkalayci N, Karadeniz EI, Elekdag-Turk S, Turk T, Cheng LL, Darendeliler MA. Effect of continuous versus intermittent orthodontic forces on root resorption: A microcomputed tomography study. Angle Orthod. 2018;88:733–9. https://doi.org/10.2319/012518-68.1.
48. Liu H, Cheema U, Player DJ. Photobiomodulation therapy (PBMT) in skeletal muscle regeneration: A comprehensive review of mechanisms, clinical applications, and future directions. Photodiagn Photodyn Ther. 2025;53:104634. https://doi.org/10.1016/j.pdpdt.2025.104634.
49. Dompe C, Moncrieff L, Matys J, Grzech-Leśniak K, Kocherova I, Bryja A, et al. Photobiomodulation—Underlying Mechanism and Clinical Applications. JCM. 2020;9:1724. https://doi.org/10.3390/jcm9061724.
50. Hamblin MR. Photobiomodulation or low-level laser therapy. J Biophotonics. 2016;9:1122–4. https://doi.org/10.1002/jbio.201670113.
51. Yong J, Gröger S, Von Bremen J, Martins Marques M, Braun A, Chen X, et al. Photobiomodulation therapy assisted orthodontic tooth movement: potential implications, challenges, and new perspectives. J Zhejiang Univ Sci B. 2023;24:957–73. https://doi.org/10.1631/jzus.B2200706.
52. Ozturk T, Gul Amuk N. Effects of photobiomodulation at different wavelengths on orthodontically induced root resorption in orthodontic retention period: a micro-CT and RT-PCR study. Lasers Med Sci. 2020;35:1419–29. https://doi.org/10.1007/s10103-020-03014-1.
53. Amaroli A, Ravera S, Baldini F, Benedicenti S, Panfoli I, Vergani L. Photobiomodulation with 808-nm diode laser light promotes wound healing of human endothelial cells through increased reactive oxygen species production stimulating mitochondrial oxidative phosphorylation. Lasers Med Sci. 2019;34:495–504. https://doi.org/10.1007/s10103-018-2623-5.
54. Gholami L, Khorsandi K, Fekrazad R. Effect of red and near-infrared irradiation on periodontal ligament stem cells: ROS generation and cell cycle analysis. J Biomol Struct Dynamics. 2023;41:10051–8. https://doi.org/10.1080/07391102.2022.2152869.
55. Kang M, Lee Y, Lee Y, Kim E, Jo J, Shin H, et al. Wavelength-dependent photobiomodulation (PBM) for proliferation and angiogenesis of melanoma tumor in vitro and in vivo. J Photochem Photobiol B. 2024;258:112990. https://doi.org/10.1016/j.jphotobiol.2024.112990.
56. Cronshaw M, Parker S, Anagnostaki E, Mylona V, Lynch E, Grootveld M. Photobiomodulation Dose Parameters in Dentistry: A Systematic Review and Meta-Analysis. Dentistry J. 2020;8:114. https://doi.org/10.3390/dj8040114.
57. Hlaing EEH, Ishihara Y, Wang Z, Odagaki N, Kamioka H. Role of intracellular Ca^{2+} –based mechanotransduction of human periodontal ligament fibroblasts. FASEB J. 2019;33:10409–24. https://doi.org/10.1096/fj.201900484R.
58. Díaz-Piña DA, Rivera-Ramírez N, García-López G, Díaz NF, Molina-Hernández A. Calcium and Neural Stem Cell Proliferation. IJMS. 2024;25:4073. https://doi.org/10.3390/ijms25074073.
59. Pascoal S, Oliveira S, Ascione M, Pereira J, Carvalho Ó, Pinho T. Effects of Vibration on Accelerating Orthodontic Tooth Movement in Clinical and In Vivo Studies: A Systematic Review. Dentistry J. 2024;12:243. https://doi.org/10.3390/dj12080243.
60. Alikhani M, Alansari S, Hamidaddin MA, Sangsuwon C, Alyami B, Thirumoorthy SN, et al. Vibration paradox in orthodontics: Anabolic and catabolic effects. PLoS ONE. 2018;13:e0196540. https://doi.org/10.1371/journal.pone.0196540.
61. Xia P, Wang X, Qu Y, Lin Q, Cheng K, Gao M, et al. TGF-β1-induced chondrogenesis of bone marrow mesenchymal stem cells is promoted by low-intensity pulsed ultrasound through the integrin-mTOR signaling pathway. Stem Cell Res Ther. 2017;8:281. https://doi.org/10.1186/s13287-017-0733-9.
62. Yan X, Han H, Zhang S, Lu Y, Ren L, Tang Y, et al. N/OFQ modulates orofacial pain induced by tooth movement through CGRP-dependent pathways. BMC Neurosci. 2021;22:25. https://doi.org/10.1186/s12868-021-00632-5.
63. Sakamoto M, Fukunaga T, Sasaki K, Seiryu M, Yoshizawa M, Takeshita N, et al. Vibration enhances osteoclastogenesis by inducing RANKL expression via NF-κB signaling in osteocytes. Bone. 2019;123:56–66. https://doi.org/10.1016/j.bone.2019.03.024.
64. Sun T, Yan Z, Cai J, Shao X, Wang D, Ding Y, et al. Effects of mechanical vibration on cell morphology, proliferation, apoptosis, and cytokine expression/secretion in osteocyte-like MLO-Y4 cells exposed to high glucose. Cell Biol Int. 2020;44:216–28. https://doi.org/10.1002/cbin.11221.
65. Siriphan N, Leethanakul C, Thongudomporn U. Effects of two frequencies of vibration on the maxillary canine distalization rate and RANKL and OPG secretion: A randomized controlled trial. Orthod Craniofac Res. 2019;22:131–8. https://doi.org/10.1111/ocr.12301.

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